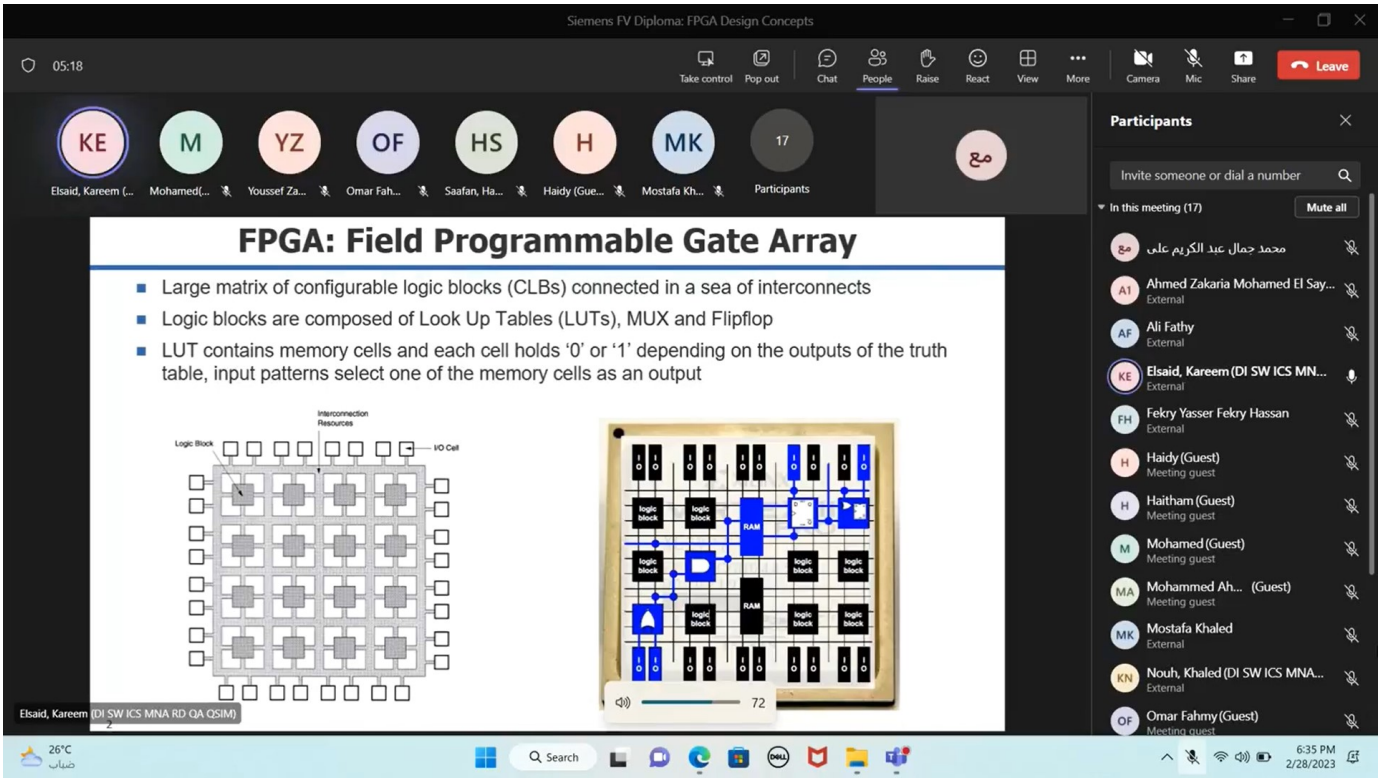
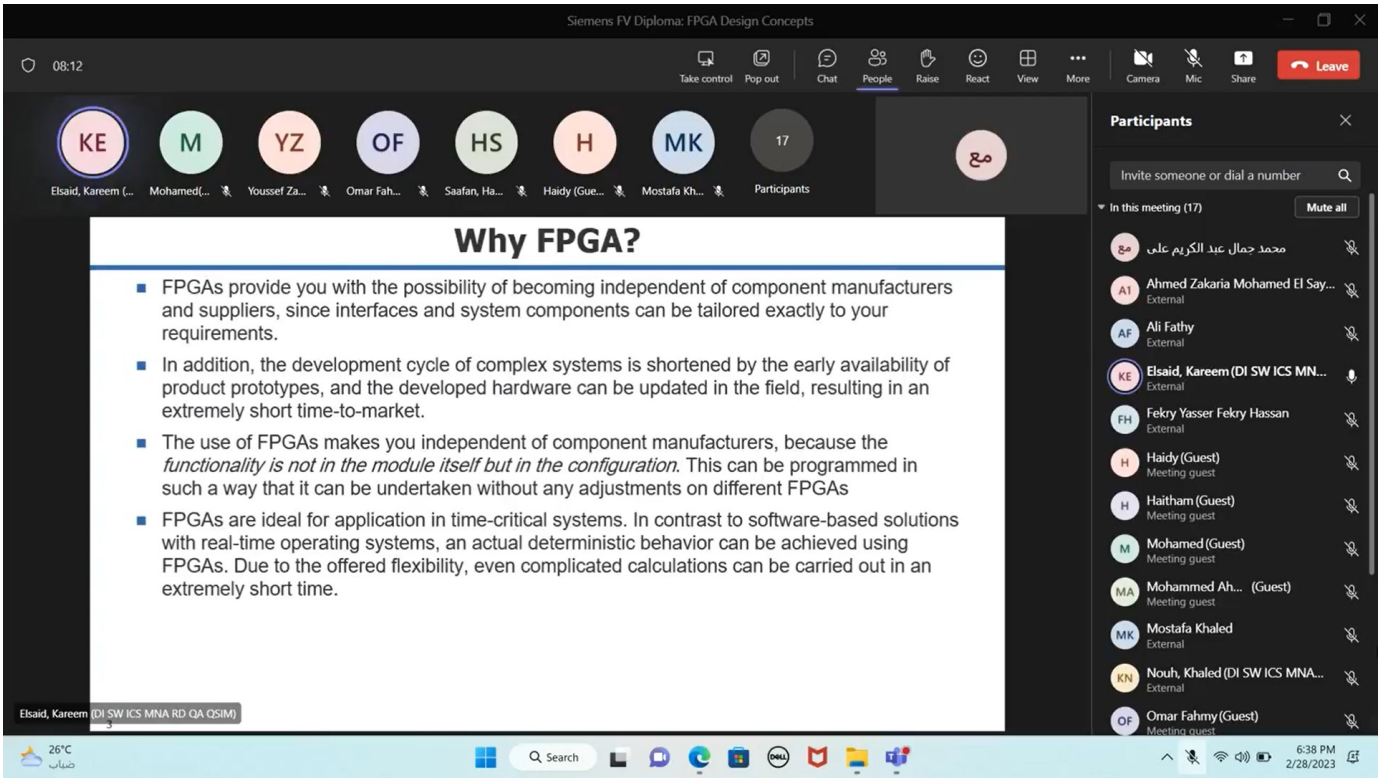
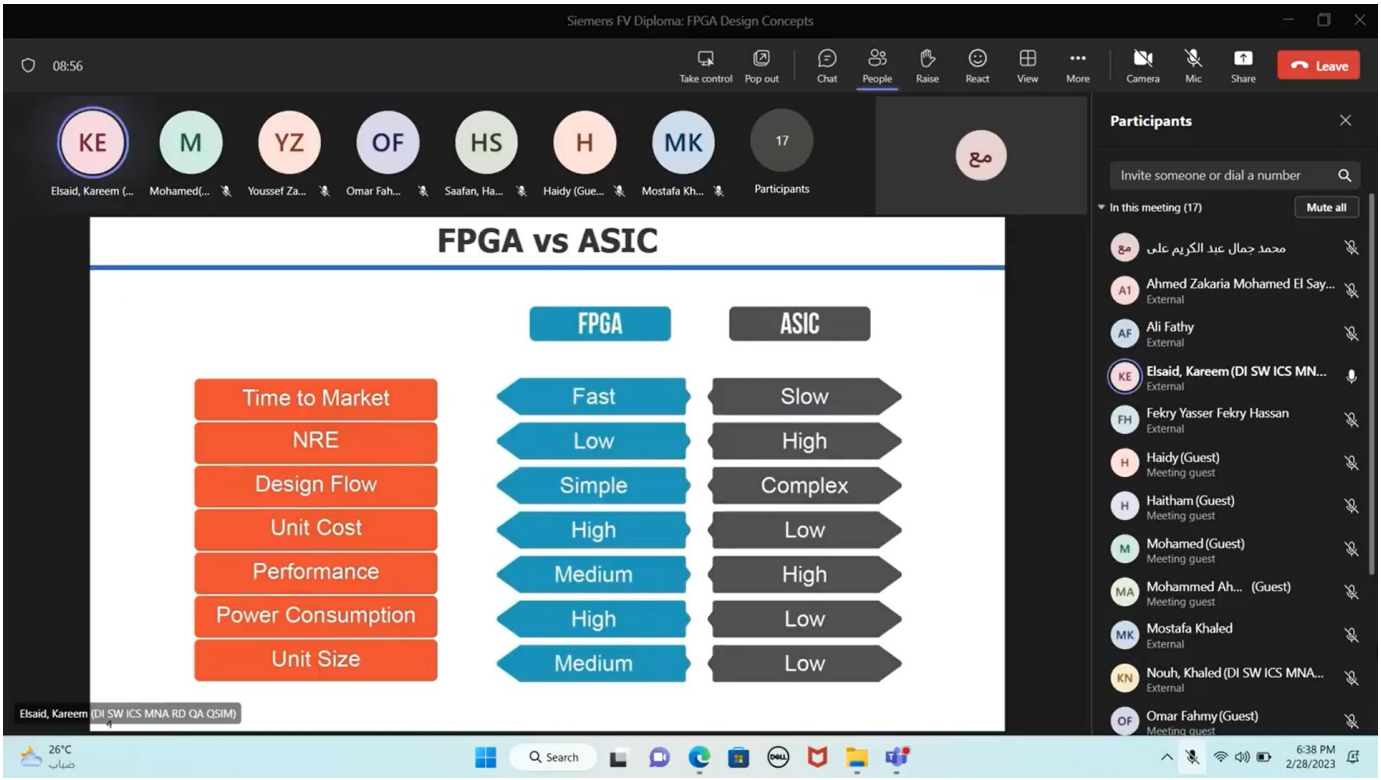


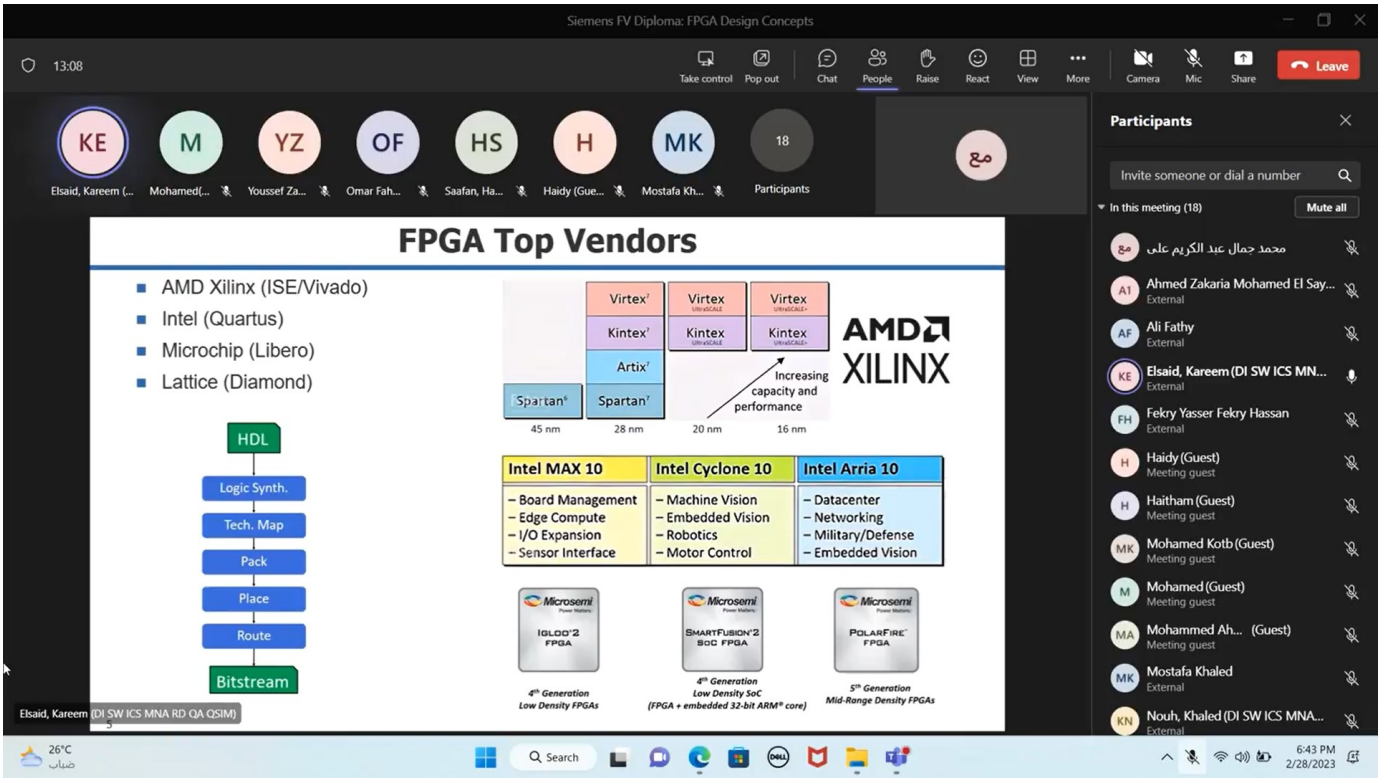
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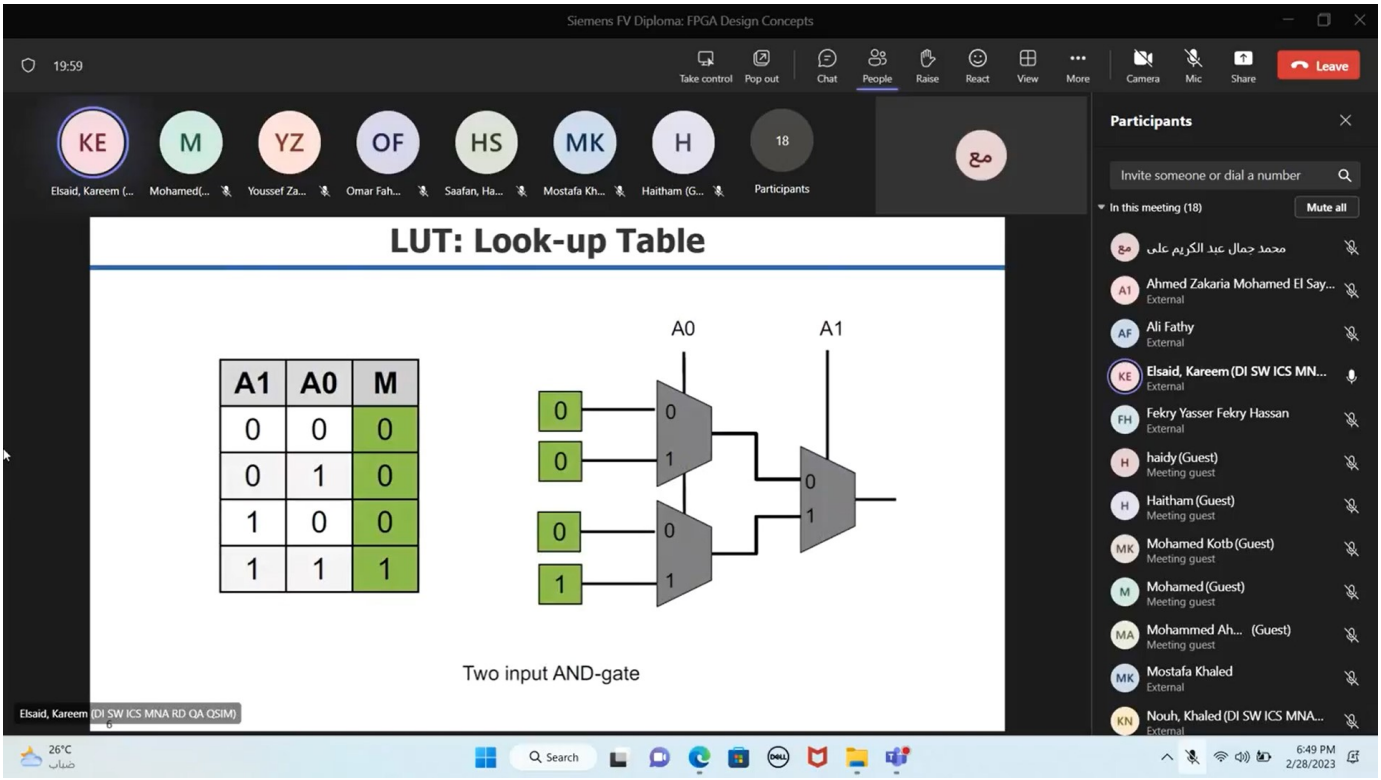
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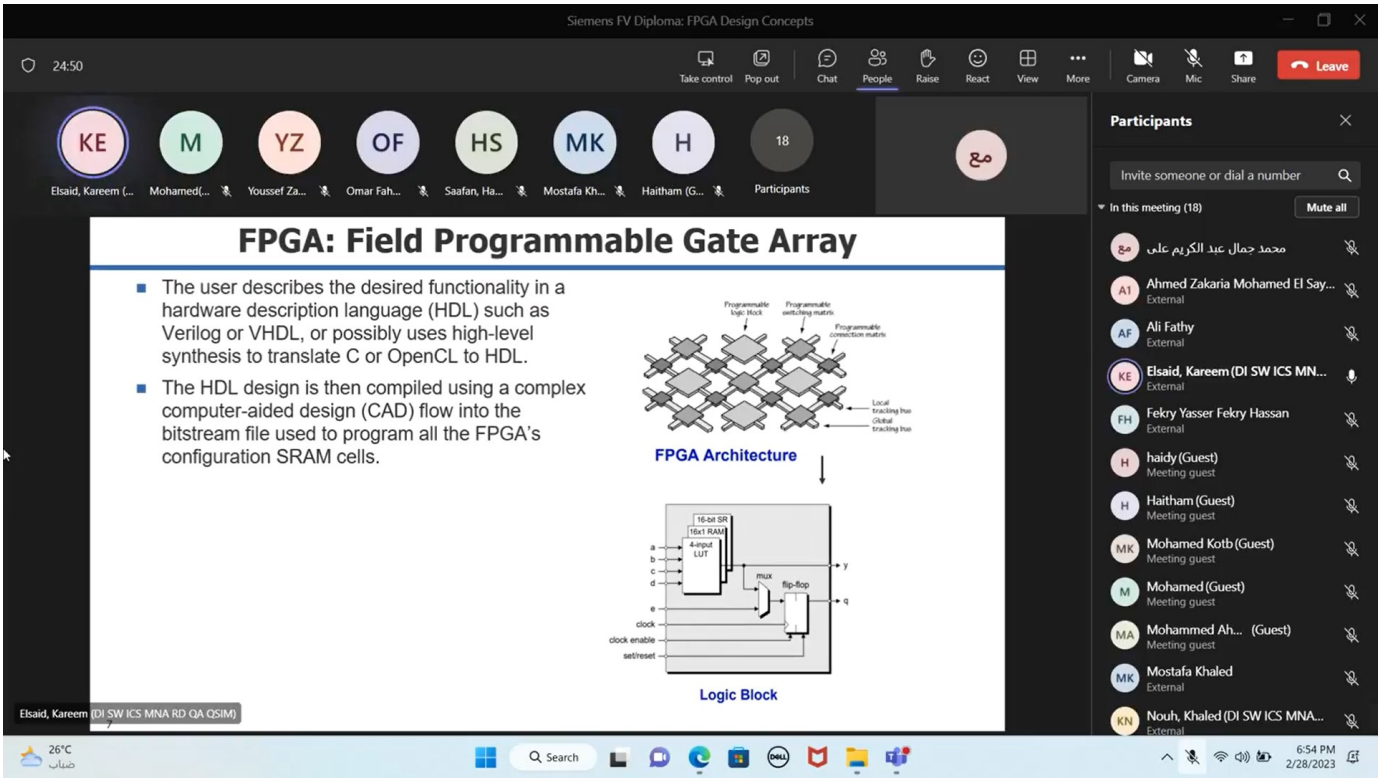
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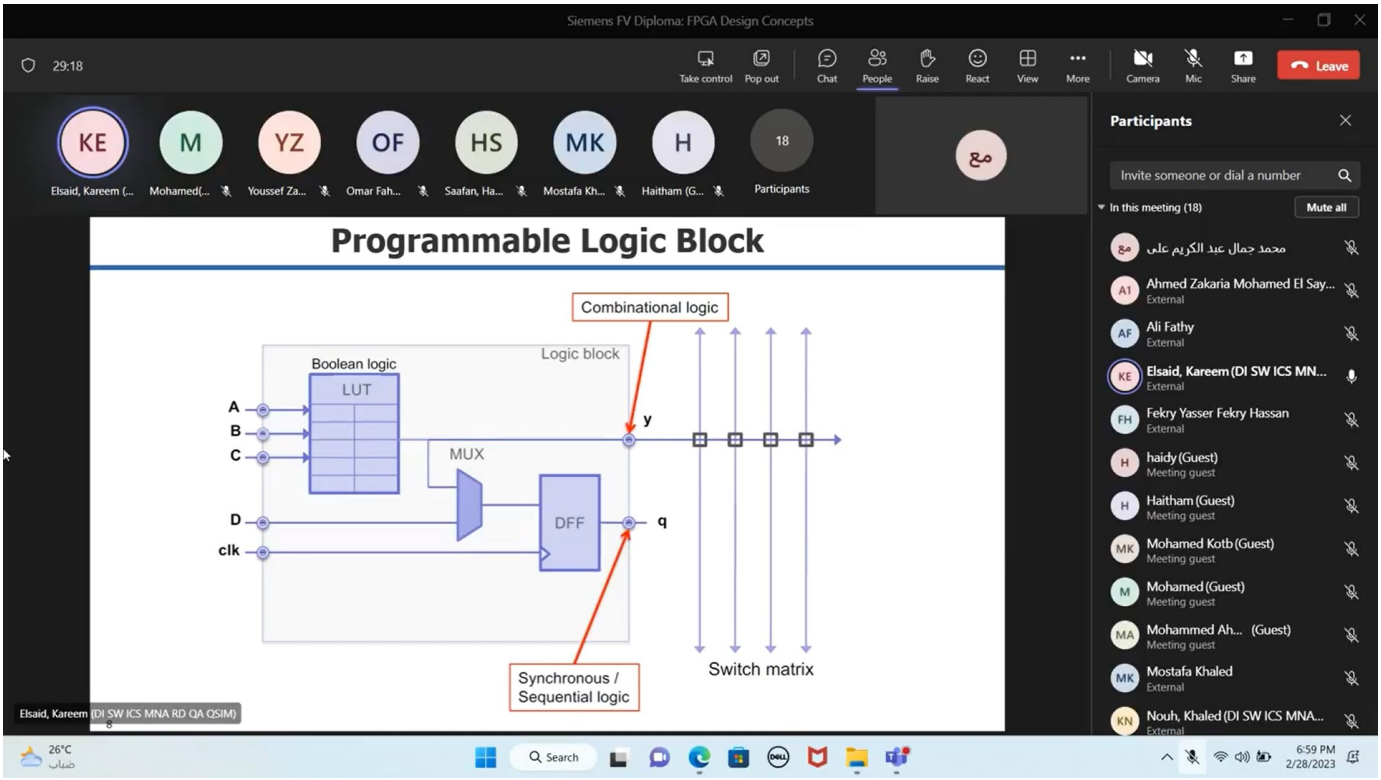
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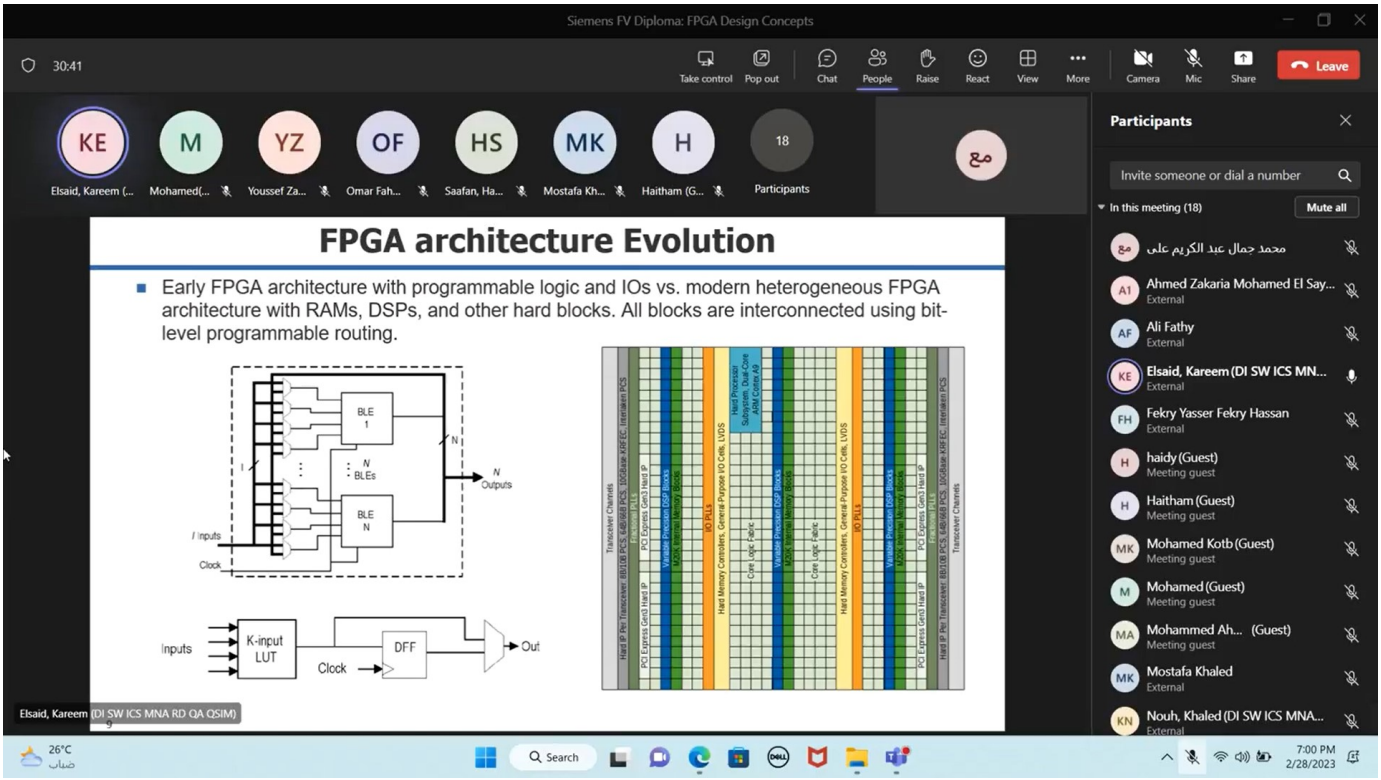
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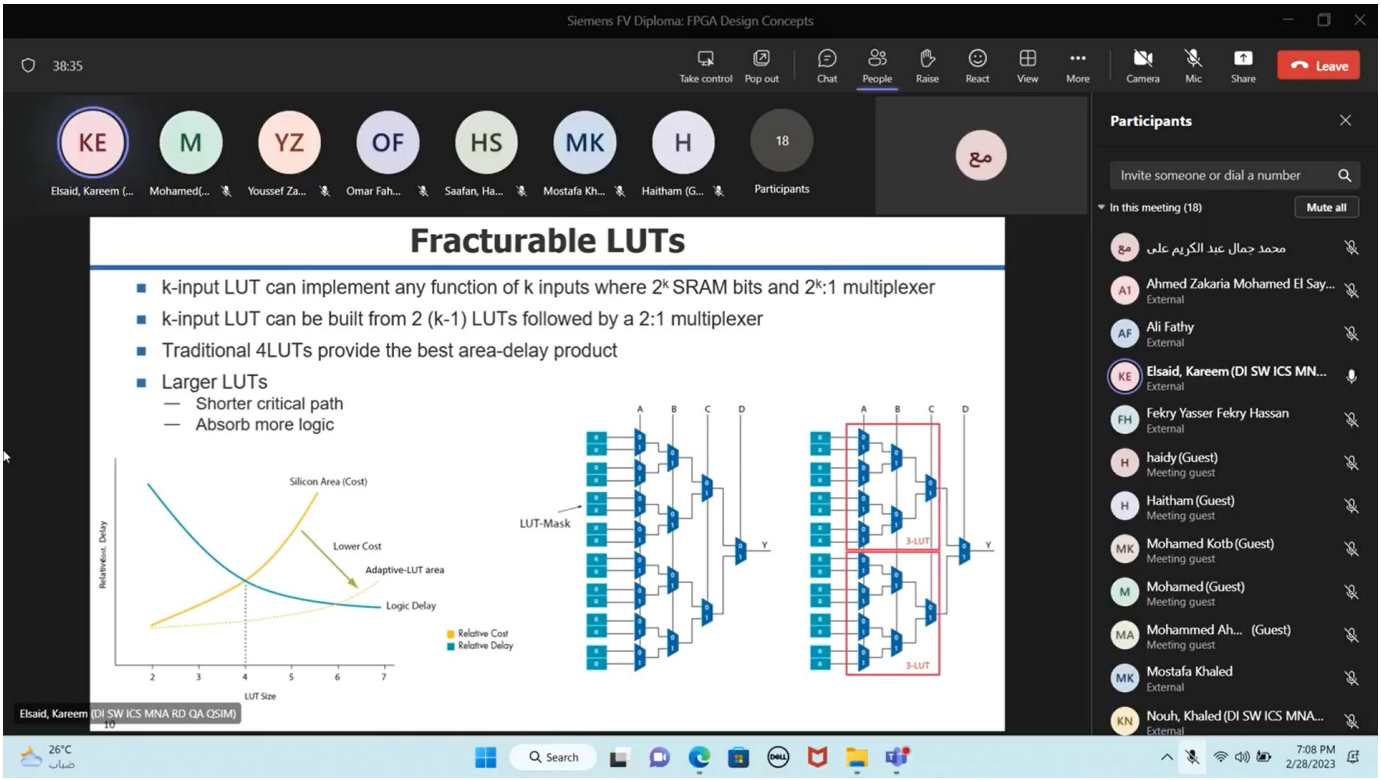
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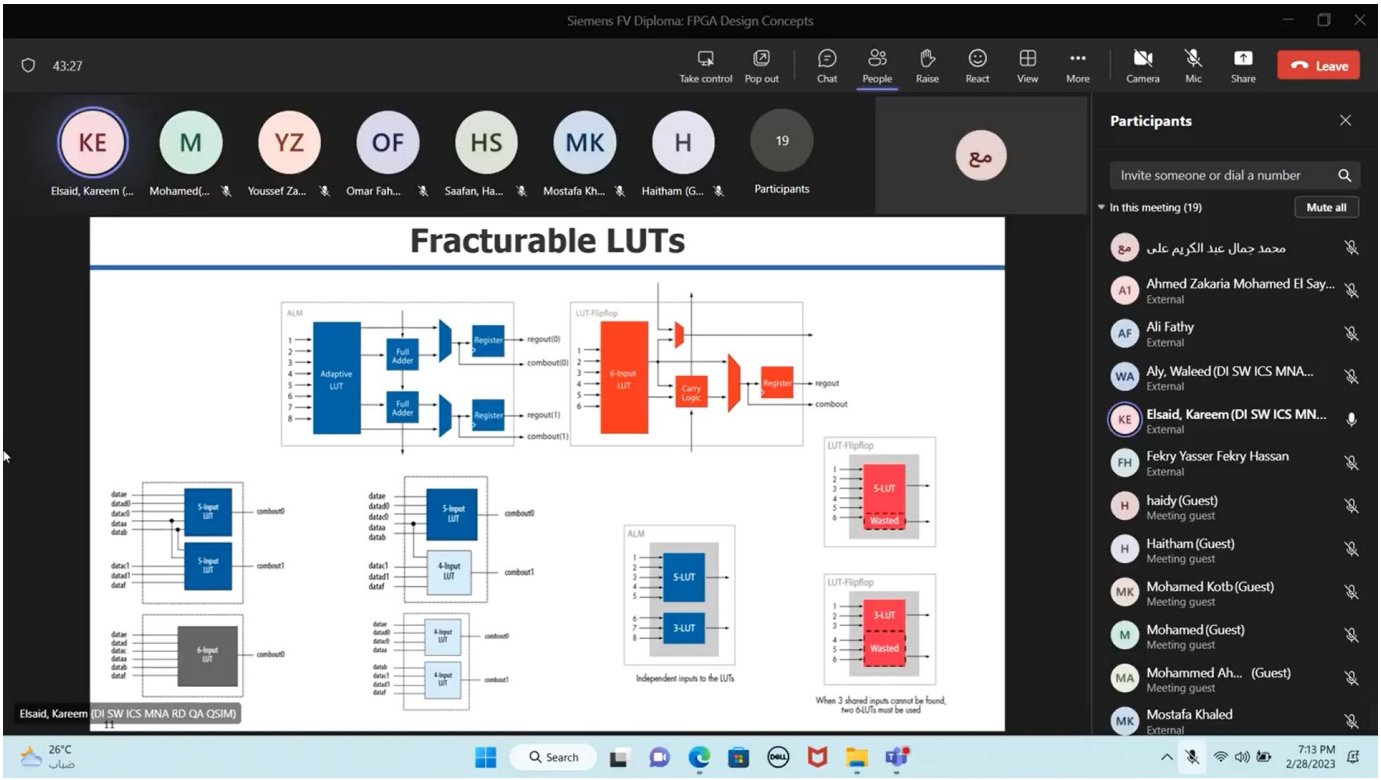
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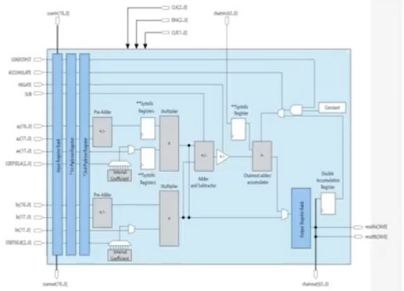
Take control | Pop out | Chat | People | Raise | React | View | More | Camera | Mic | Share | Leave

KE Elsaid, Kareem (... | M Mohamed(... | OF Omar Fah... | MK Mostafa Kh... | H Haitham (G... | مم مريم علي عـ... | FH Fekry Yasse... | 18 Participants | مع

DSP Blocks: Digital Signal Processing Blocks

- Only carry chains were used to implement efficient adders as dedicated arithmetic circuits in commercial FPGA architectures.
- As a result, carry chains and LUTs had to be used to implement multipliers in the soft logic, resulting in a sizable area and delay penalty.

DSP Implementations	Block Architecture
Fixed-Point Arithmetic	• Input register bank • Pipeline register • Pre-adder/subtract • Internal coefficient • Multipliers • Adder and Subtractor • Accumulator, chainout adder, and Preload Constant • Systolic registers • Double accumulation register • Output register bank
Floating-Point Arithmetic	• Input register bank • Pipeline register • Multipliers • Adder • Accumulator • Output register bank • Exception Handling



Elsaid, Kareem (DI SW ICS MNA RD QA QSIM)

12

Participants

Invite someone or dial a number

In this meeting (18) | Mute all

- مع محمد جمال عبد الكريم على
- A1 Ahmed Zakaria Mohamed El Say... External
- AF Ali Fathy External
- WA Aly, Waleed (DI SW ICS MNA... External
- KE Elsaid, Kareem (DI SW ICS MN... External
- FH Fekry Yasser Fekry Hassan External
- H haidy (Guest) Meeting guest
- H Haitham (Guest) Meeting guest
- MK Mohamed Kotb (Guest) Meeting guest
- M Mohamed (Guest) Meeting guest
- MA Mohammed Ah... (Guest) Meeting guest
- MK Mostafa Khaled External

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Inferring arithmetic blocks

Example 3: 18x18-Bit Signed Multiplier with Registered I/Os

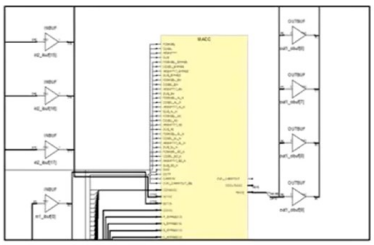
This is code for an 18x18 signed multiplier. The inputs and outputs are registered, with a synchronous active low reset signal. The synthesis tool fits all this logic into one MACC block, as shown below.

```
module sign18x18_mult ( in1, in2, clk, rst, out1 );

input signed [17:0] in1, in2;
input clk;
input rst;
output signed [40:0] out1;
reg signed [40:0] out1;
reg signed [17:0] in1_reg, in2_reg;

always @ ( posedge clk )
begin
   if ( ~rst )
      begin
         in1_reg <= 18'b0;
         in2_reg <= 18'b0;
         out1 <= 41'b0;
      end
      else
      begin
         in1_reg <= in1;
         in2_reg <= in2;
         out1 <= in1_reg * in2_reg;
      end
end

endmodule
```



Elsaid, Kareem (DI SW ICS MNA RD QA QSIM)

Participants

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In this meeting (18) | Mute all

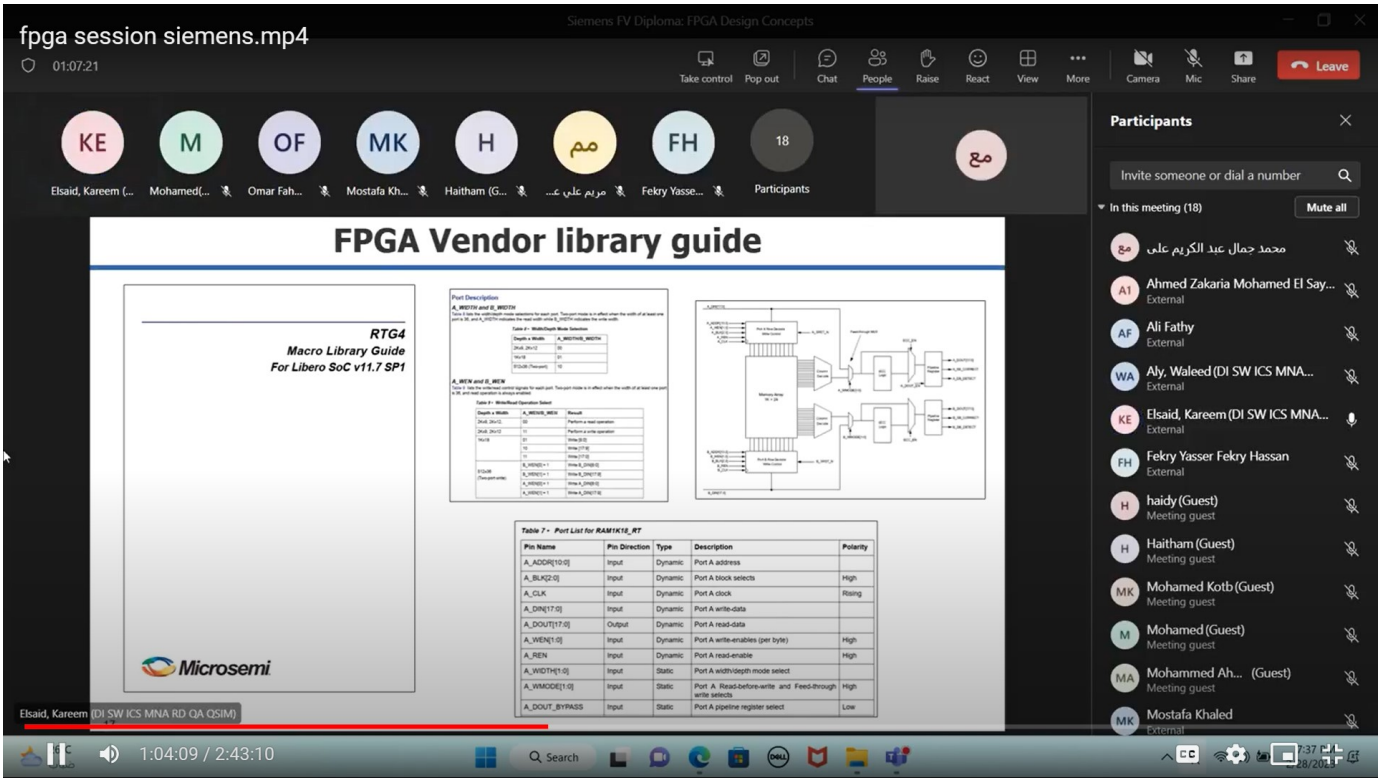
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- Ahmed Zakaria Mohamed El Say... External
- Ali Fathy External
- Aly, Waleed (DI SW ICS MNA... External
- Elsaid, Kareem (DI SW ICS MN... External
- Fekry Yasser Fekry Hassan External
- haidy (Guest) Meeting guest
- Haitham (Guest) Meeting guest
- Mohamed Kotb (Guest) Meeting guest
- Mohamed (Guest) Meeting guest
- Mohammed Ah... (Guest) Meeting guest
- Mostafa Khaled External

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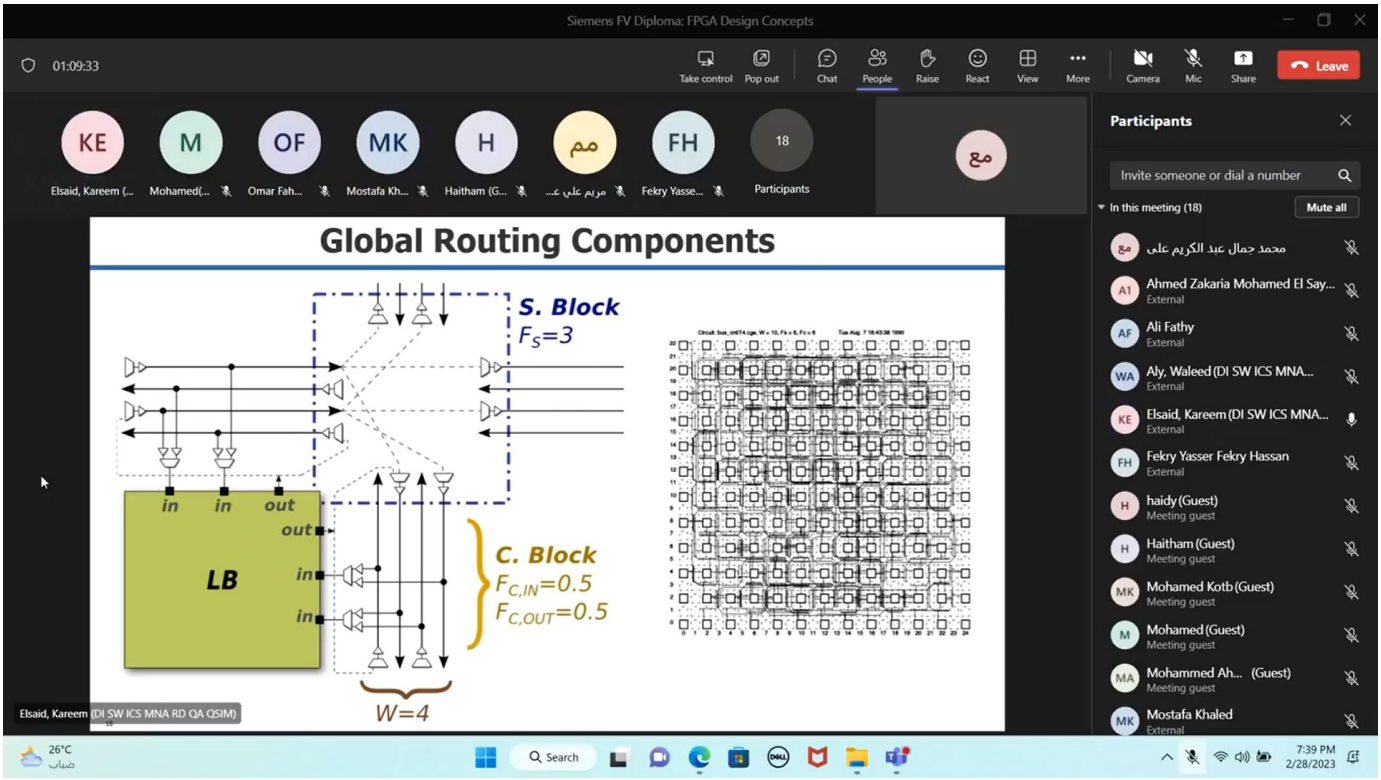
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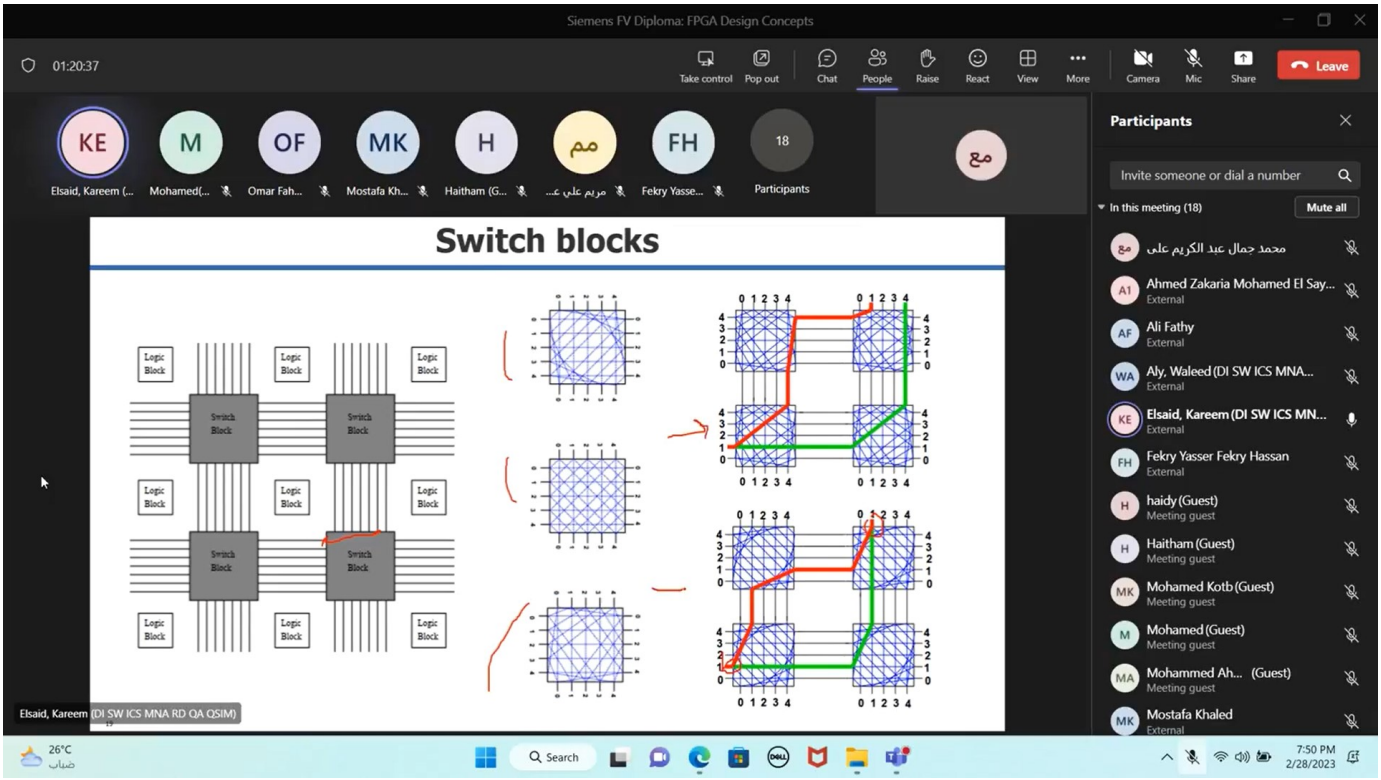
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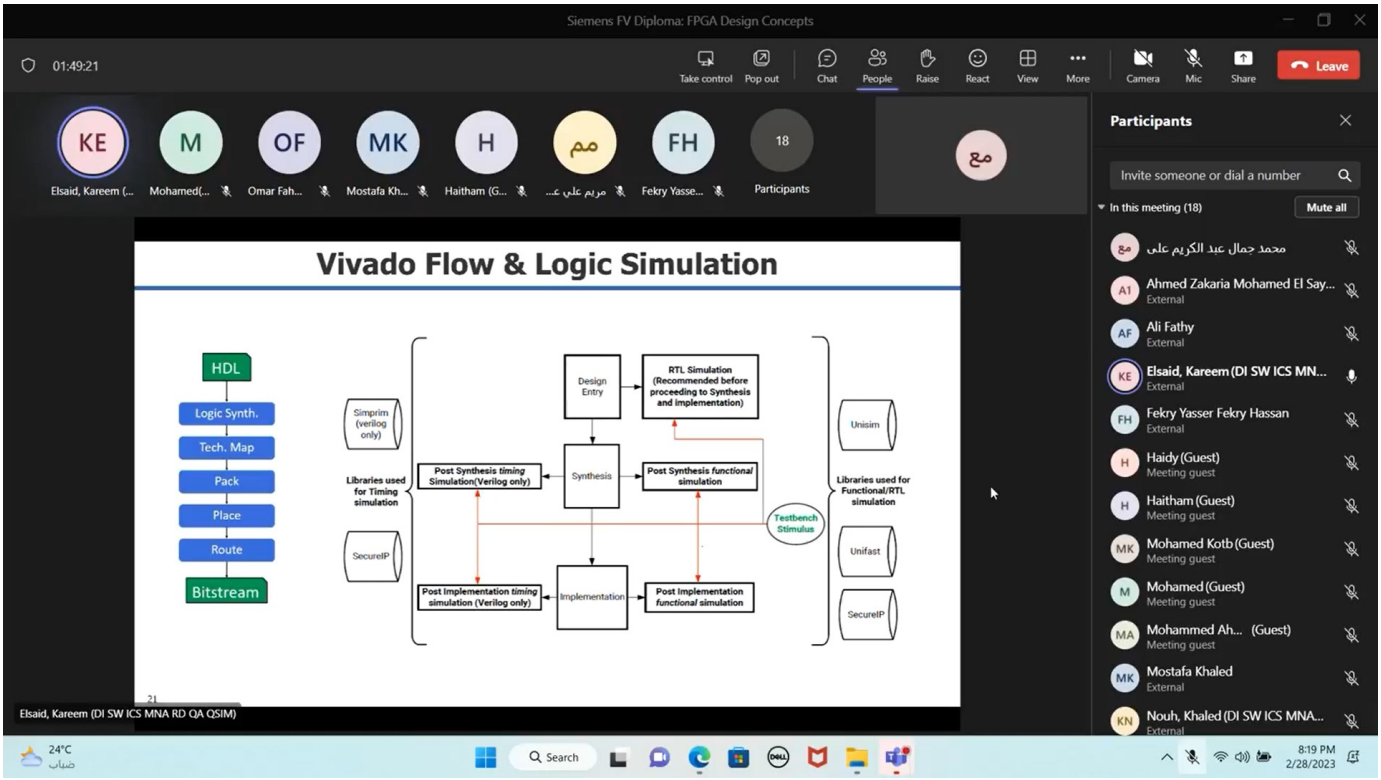
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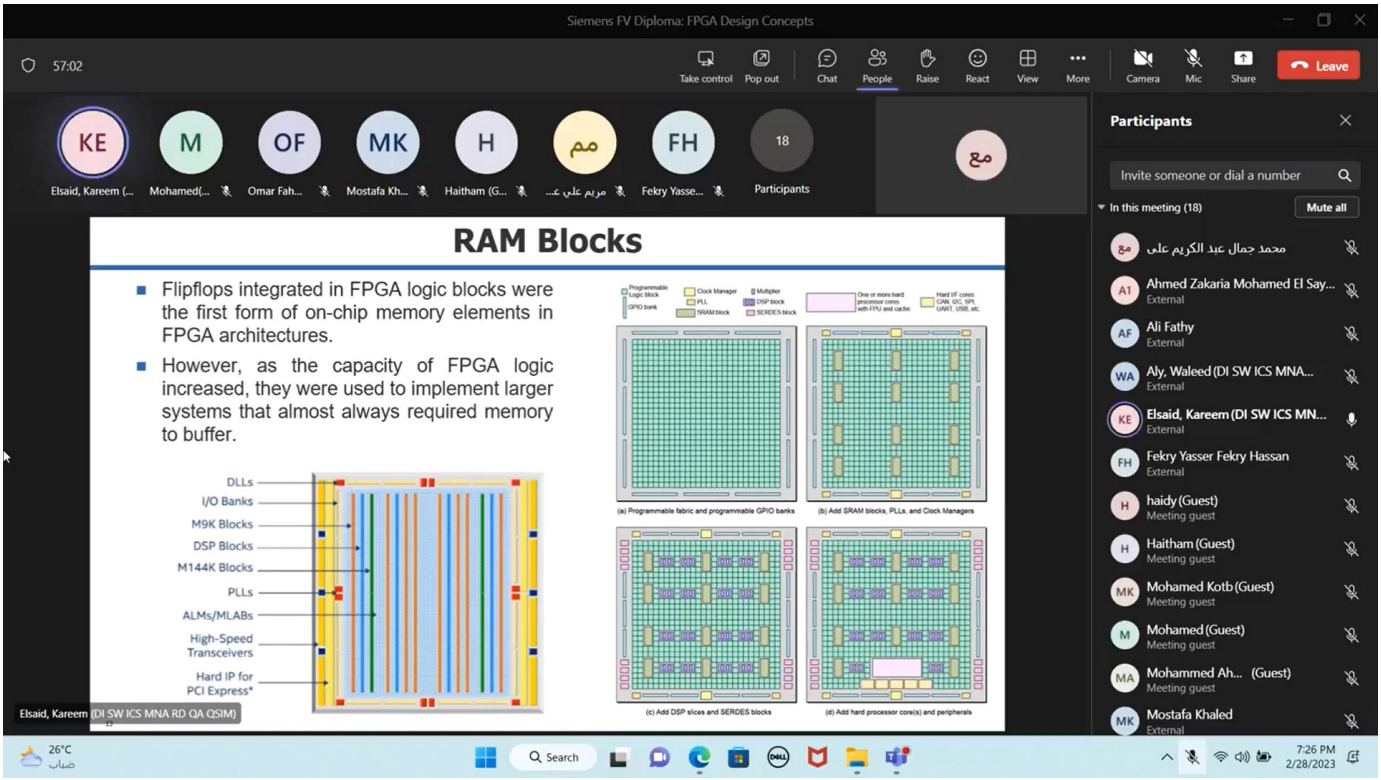
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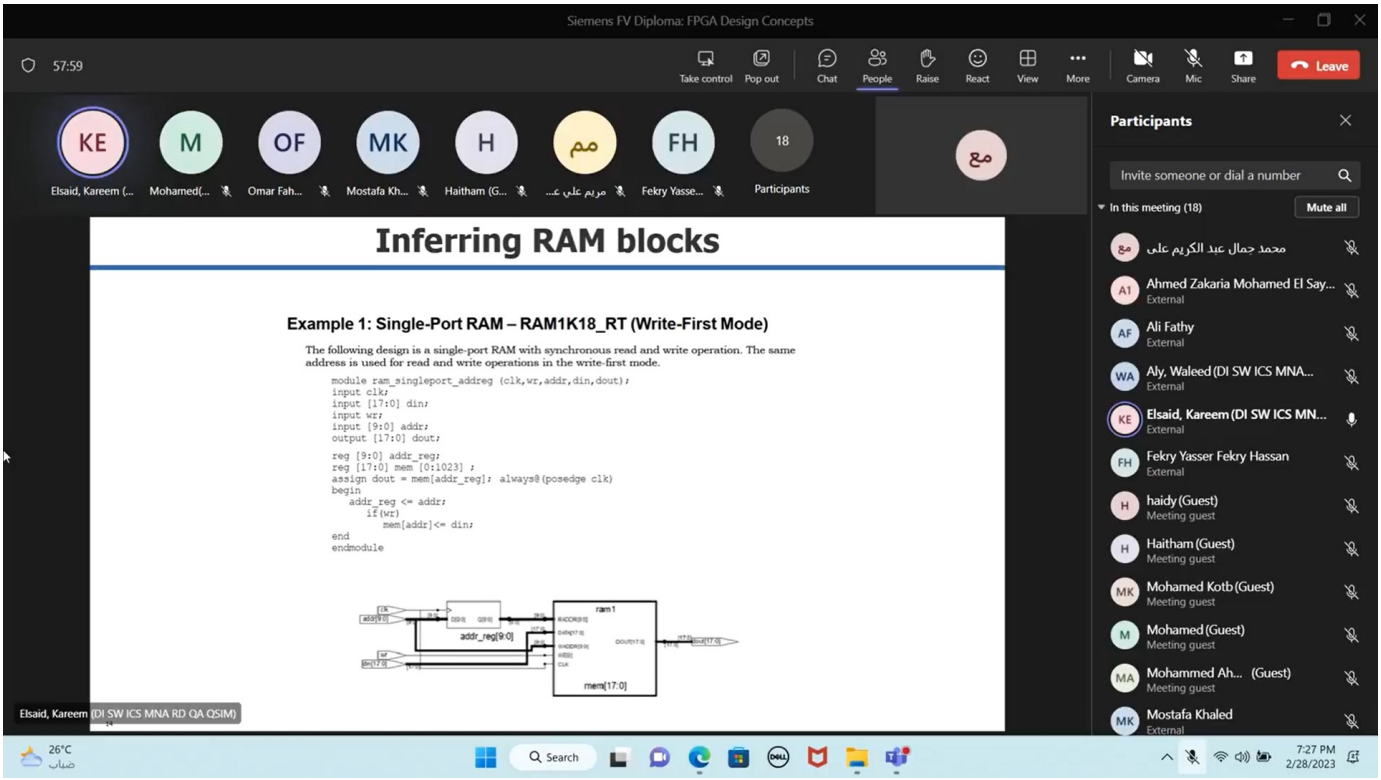
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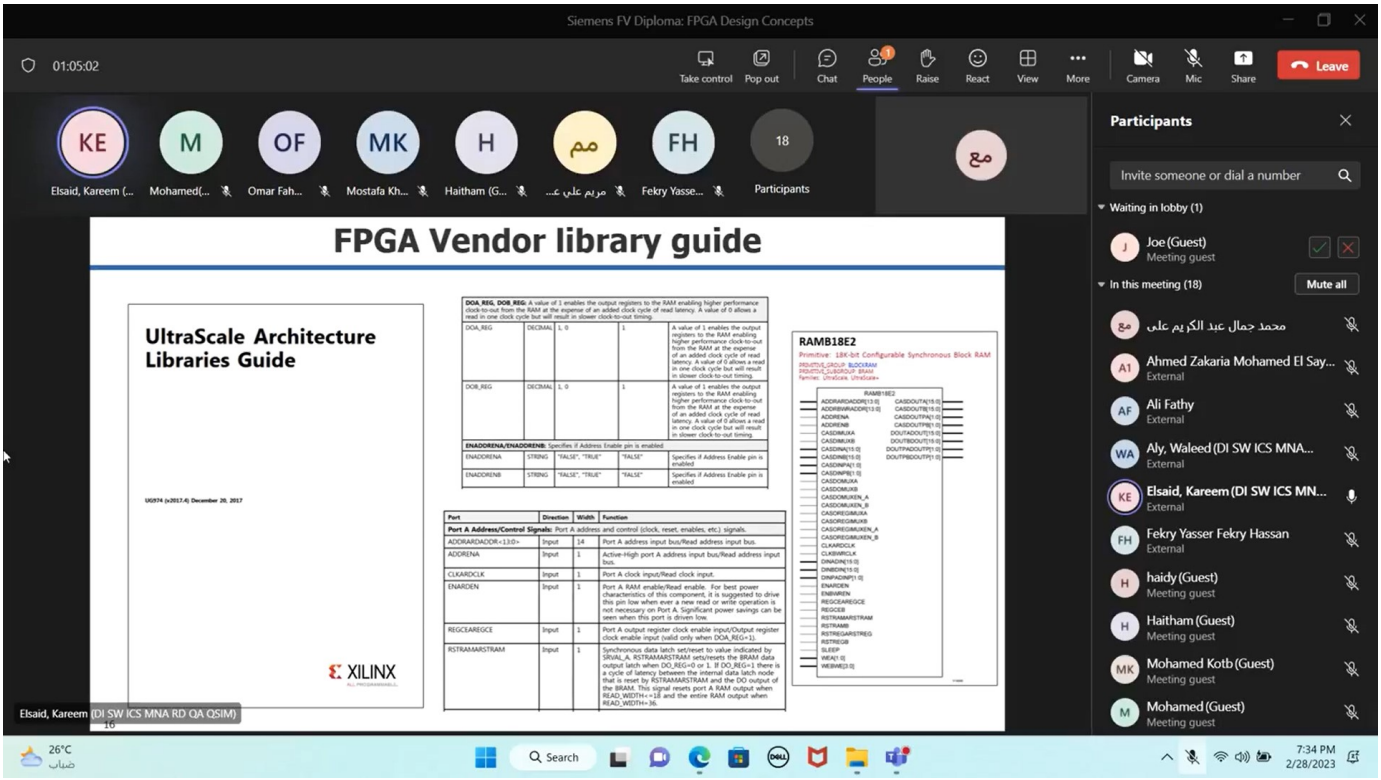
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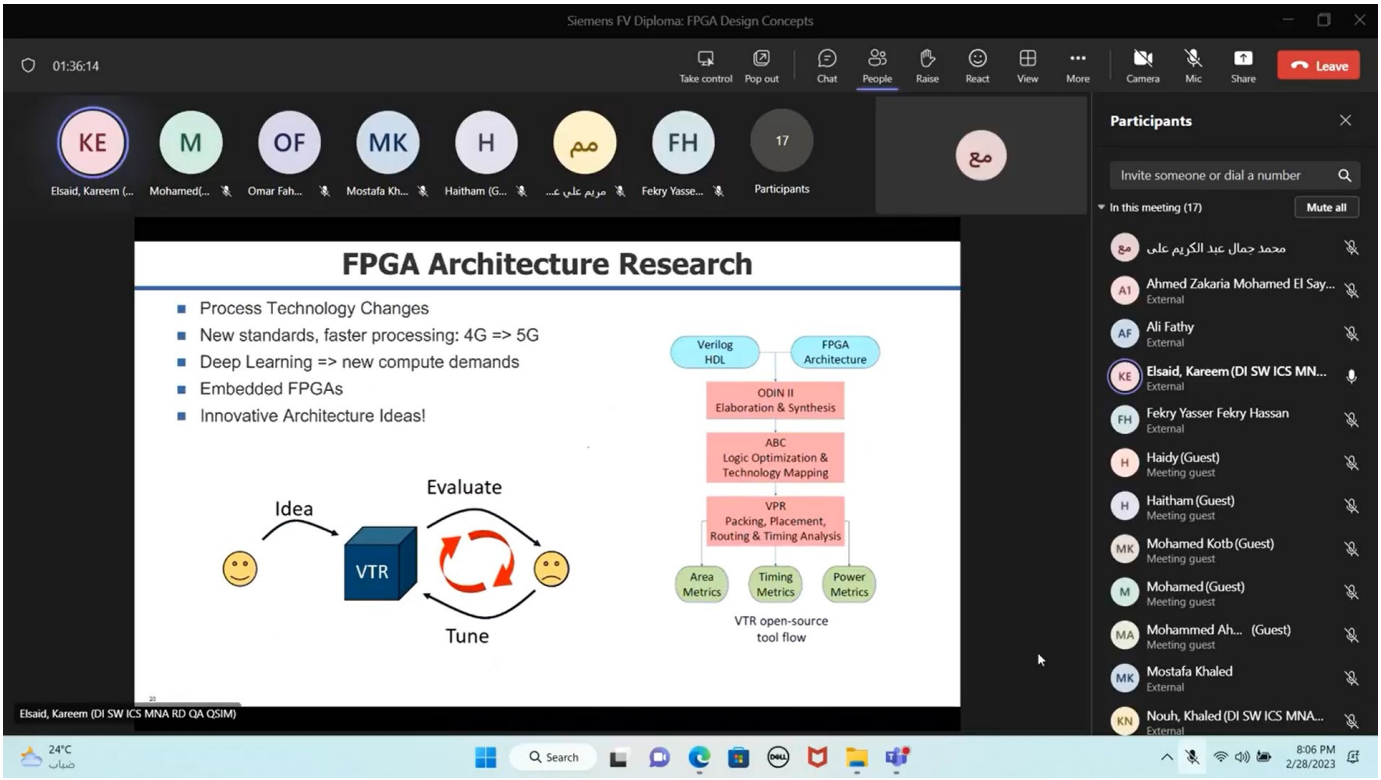
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